

# GARP Risk & AI — 250 Detailed Hard Questions

Unofficial Exam Prep

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## Question

A tier-1 bank has deployed a deep neural network to screen consumer loan applications. When a rejected applicant files a formal complaint under GDPR's "right to explanation", the compliance team asks the data science team for a justification. The data science team can run a SHAP explainer that produces the top three features driving the specific rejection, but when asked to describe how the model internally transforms the input vector into the output probability, they admit that not even they can trace this path. The regulator accepts the SHAP output as a valid post-hoc explanation. Which statement most accurately characterises this scenario?

- A. The model is interpretable but not explainable, because SHAP values are a property of the model rather than the specific decision.
- B. The model is explainable but not interpretable, because post-hoc explanation of a specific decision is possible even though the internal mechanism cannot be described.
- C. The model is neither explainable nor interpretable, because SHAP is a post-hoc technique that does not reveal the model's true behaviour.
- D. The model is both fully interpretable and explainable, because SHAP output is equivalent to understanding the model's internal mechanism.

Answer: B

## Explanation

Explainability refers to the ability to provide a human-understandable account of a specific model decision — typically post-hoc, using techniques like SHAP or LIME. Interpretability refers to the ability to understand the model's internal logic and mechanisms. Deep neural networks are frequently explainable but not interpretable. Option A is wrong because SHAP does not make the model internally interpretable. Option C is wrong because post-hoc explanation *does* count as explainability. Option D is wrong because SHAP values do not reveal the model's internal mechanism; they only attribute the output to inputs.

## Question

The risk management committee at a global insurance group is reviewing the group's AI inventory. The CRO notes that one class of models in the fraud-detection stack is described in the documentation as "inscrutable". A junior analyst asks what this means in the context of the GARP Risk and AI curriculum. The senior data scientist explains that inscrutability refers to a specific limitation of certain AI models. Which of the following most accurately captures the meaning of inscrutability as used in the GARP curriculum?

- A. Inscrutability refers to the inability of the model to generalise to out-of-sample data.
- B. Inscrutability refers to the inability to understand the internal mechanisms through which the model produces its outputs.
- C. Inscrutability refers to the inability to explain a specific prediction to a customer or regulator.
- D. Inscrutability refers to the inability to measure the model's accuracy with traditional statistical metrics.

Answer: B

## Explanation

Inscrutability is specifically the property of a model whose internal mechanisms are opaque — the analyst cannot trace how the input vector is transformed into the output. Option A confuses inscrutability with generalisability. Option C confuses it with explainability, which is about specific decisions. Option D confuses it with accuracy or measurability.

## Question

An asset manager is considering two ways to build a market-regime classifier. Approach One is a rule-based system that uses a fixed set of hand-coded conditions on moving averages and volatility thresholds to label regimes as “risk-on” or “risk-off”. Approach Two trains a deep neural network with several hidden layers on historical features to output the same classification. The CIO asks you which approach corresponds to which paradigm in AI and what the key trade-offs are. Which statement best captures the distinction?

- A. Approach One is deep learning and Approach Two is classical AI, because deep neural networks rely on pre-defined rules while rule-based systems learn from data.
- B. Approach One is classical AI and Approach Two is deep learning; the classical approach scales well to complex unstructured data, while the deep learning approach does not.
- C. Approach One is classical AI (rules and search within a well-defined context) and Approach Two is deep learning (learns representations via backpropagation); the deep learning approach can capture non-linear patterns but is typically more inscrutable.
- D. Both approaches are classical AI because both encode a mapping from inputs to outputs that is fixed after deployment.

Answer: C

## Explanation

Classical AI (GOFAI) uses hand-crafted rules and symbolic reasoning. Deep learning learns from data using backpropagation and gradient descent, allowing it to capture non-linear relationships, but the internal mechanism is opaque (inscrutable). Option A reverses the definitions. Option B also reverses the scalability argument. Option D ignores the fact that deep learning learns its parameters from data, which is definitionally distinct from hand-coded rules.

## Question

A history-of-AI seminar traces three milestones: (i) Deep Blue's 1997 defeat of Kasparov, (ii) AlexNet's 2012 victory in the ImageNet competition, and (iii) the 2017 publication of "Attention Is All You Need". A candidate is asked to associate each milestone with the correct AI paradigm. Which of the following correctly maps the milestones?

- A. (i) deep learning, (ii) classical AI, (iii) classical AI.
- B. (i) classical AI, (ii) deep learning, (iii) foundational work for modern LLMs via the transformer architecture.
- C. (i) classical AI, (ii) reinforcement learning, (iii) rule-based expert systems.
- D. (i) deep learning, (ii) transformer-based models, (iii) reinforcement learning.

Answer: B

## Explanation

Deep Blue was a classical AI system based on search and hand-crafted evaluation. AlexNet is a landmark deep convolutional neural network that won ImageNet in 2012, launching the modern deep learning era. "Attention Is All You Need" introduced the transformer, which underpins modern large language models. Option A and D misattribute. Option C mislabels AlexNet and the transformer paper.

## Question

A GARP-qualified risk officer is classifying AI incidents on the group's register. The following four incidents have been logged. Which incident should be classified as a *societal risk* rather than as an individual or organisational risk?

- A. A trading algorithm at a mid-sized broker transmits an erroneous order that causes USD 20 million of client losses.
- B. A hospital's AI triage tool mis-prioritises a patient's case, delaying treatment by several hours.
- C. A large social media platform's recommendation algorithm is shown to systematically amplify divisive political content, contributing to measurable erosion in public trust in democratic institutions across several countries.
- D. A wealth management firm's AI onboarding tool fails to detect a sophisticated identity-fraud pattern, resulting in a small number of account takeovers.

Answer: C

## Explanation

Societal risk refers to harm to society at scale — such as misinformation, erosion of trust in institutions, or damage to democratic processes. Option A is an organisational risk (client losses and reputational exposure for the firm). Option B is individual risk (harm to a specific patient). Option D is individual plus organisational (specific customers harmed, specific firm exposed). Only C clearly describes harm at the societal level.

## Question

A risk analyst observes that a hedge fund's new AI-driven news-parsing engine routinely outperforms a legacy keyword-based NLP pipeline on the task of extracting sentiment from unstructured earnings-call transcripts. The legacy system relies on a hand-curated sentiment lexicon; the new system uses a pretrained transformer fine-tuned on the firm's transcripts. Which of the following best explains the performance gap in terms of the classical-AI versus deep-learning distinction?

- A. The legacy system fails because it cannot process numerical data, whereas the transformer can handle both text and numbers.
- B. The legacy system cannot adapt to the context in which words appear; the transformer's attention mechanism captures long-range dependencies and polysemy, allowing it to learn patterns that hand-crafted lexicons miss.
- C. The legacy system is more accurate in theory but is hampered by bugs; the transformer has no such limitation.
- D. The performance gap is due solely to the larger dataset available to the transformer, not to any architectural difference.

Answer: B

## Explanation

Classical NLP pipelines based on lexicons fail on context-dependent meaning, negation, sarcasm, and multi-word expressions. Transformer-based models use self-attention to build contextual embeddings that capture long-range dependencies and polysemy, allowing them to learn patterns that hand-crafted dictionaries miss. Option A is factually wrong. Option C is not a plausible explanation. Option D is incomplete: data matters, but the architectural difference is decisive.

## Question

A quantitative risk analyst is preparing a feature set for a credit-scoring model. The dataset contains the following five fields: (i) an internal credit rating on a scale of 1–10 from lowest to highest risk; (ii) the applicant's industry code from a list of 20 mutually exclusive industry classifications; (iii) the applicant's annual revenue in USD; (iv) the applicant's age in years; (v) the applicant's self-reported ESG score on a scale of 1–5 where 1 is "very poor" and 5 is "very good" but the intervals are not known to be equal. Which pair correctly classifies (i) and (ii)?

- A. (i) is nominal; (ii) is ordinal.
- B. (i) is ordinal; (ii) is nominal.
- C. (i) is interval; (ii) is ratio.
- D. (i) is nominal; (ii) is interval.

Answer: B

## Explanation

(i) is an internal credit rating on an ordered scale with unequal intervals between levels — this is ordinal. (ii) is an industry code with no inherent ordering between categories — this is nominal. Options A, C, and D misclassify one or both fields.

## Question

A modelling team is preparing to fit a KNN-based fraud detector on features that include “transaction amount in USD” (range 1 to 1,000,000) and “days since account opened” (range 1 to 3,650). A junior analyst proposes min-max normalization on both features. A senior analyst disagrees and proposes z-score standardization. Which of the following statements best captures the trade-off?

- A. Min-max normalization should always be preferred because it produces bounded inputs that stabilize distance computations.
- B. Standardization should be preferred when the feature distribution contains extreme outliers, because min-max normalization compresses the bulk of the distribution into a narrow sub-interval of  $[0,1]$ .
- C. Standardization should always be preferred regardless of the shape of the distribution, because it is distribution-free.
- D. Neither scaling method should be applied, because KNN is invariant to monotone transformations of the features.

Answer: B

## Explanation

Min-max normalization is sensitive to outliers because the min and max are affected by extreme values; the rest of the distribution is squeezed into a narrow band. Standardization uses mean and standard deviation, which are less sensitive to outliers, and preserves the shape of the underlying distribution. Option A is wrong because min-max is exactly the wrong choice under heavy tails. Option C is wrong because standardization is not distribution-free in effect; it is still a linear transformation. Option D is wrong because KNN is not invariant to monotone transformations; distances change.

## Question

An analyst runs forward stepwise regression on a dataset of annual salaries using polynomial terms in age. At step two, the model "Age + Age<sup>3</sup>" achieves AIC = 3123.20. At step three, the model "Age + Age<sup>3</sup> + Age<sup>5</sup>" achieves AIC = 3124.50. The analyst must report which model to select and why. Which of the following is the correct conclusion?

- A. Select "Age + Age<sup>3</sup> + Age<sup>5</sup>" because adding more terms cannot worsen the model's fit.
- B. Select "Age + Age<sup>3</sup>" because AIC penalises complexity, and the increase from 3123.20 to 3124.50 indicates that the additional term is not justified.
- C. Select "Age + Age<sup>3</sup> + Age<sup>5</sup>" because AIC is monotone in the number of parameters and only the direction of change matters.
- D. Neither model is valid because AIC cannot be compared across models with different numbers of parameters.

Answer: B

## Explanation

AIC explicitly penalises additional parameters. In forward stepwise regression, we add a term only if it reduces AIC. Since adding Age<sup>5</sup> increased AIC from 3123.20 to 3124.50, the term is not justified. Option A is wrong because AIC, unlike R<sup>2</sup>, can worsen when unnecessary terms are added. Option C confuses AIC with R<sup>2</sup>. Option D is wrong because AIC is designed for cross-model comparison.

## Question

A research team is asked to test the hypothesis that the CAPM beta of a particular equity is significantly different from one. Separately, another team is asked to build a model that maximises the out-of-sample prediction accuracy of next-day returns for the same equity. Which of the following statements correctly matches each task to the appropriate methodology?

- A. Both tasks should use machine learning, because ML is always superior when the data is large.
- B. Both tasks should use classical statistics, because ML cannot handle financial time-series data.
- C. The first task is best served by classical statistics (inference, t-test on  $\beta$ ); the second task is best served by machine learning (regularised regression, tree ensembles, or neural networks optimised for out-of-sample prediction).
- D. The first task is best served by machine learning; the second by classical statistics.

Answer: C

## Explanation

The first task is a hypothesis test — this is inference, the domain of classical statistics. The second task is a prediction problem — this is the domain of machine learning. Option A is wrong because ML's strengths lie in prediction, not inference. Option B is wrong because ML handles time-series data well when used appropriately. Option D reverses the correct mapping.

## Question

A junior quant trains a KNN classifier on a dataset with two features: “customer tenure in months” (values between 1 and 240) and “monthly income in USD” (values between 800 and 200,000). Without scaling, the classifier’s predictions are almost entirely driven by income. She then applies z-score standardization to both features and the model performance improves dramatically. Which of the following best explains what happened?

- A. The KNN algorithm was broken before scaling; standardization fixed a bug.
- B. Because KNN uses a distance metric, features with larger numerical ranges dominate the distance computation; standardization brings both features onto a comparable scale so that tenure can contribute appropriately.
- C. Standardization increased the amount of information in the dataset.
- D. Standardization eliminated outliers that were causing the poor performance.

Answer: B

## Explanation

Distance-based algorithms (KNN, K-means, SVM, PCA) are sensitive to feature scale. A feature measured in the tens of thousands (income) dominates one measured in the tens or hundreds (tenure). Standardization puts both features on the same scale (mean zero, SD one), allowing tenure to contribute. Option A is wrong — the algorithm was not broken. Option C is wrong — scaling does not add information. Option D is wrong — standardization does not remove outliers; it merely scales.

## Question

An analyst estimates a linear regression of salary on experience plus dummy variables for three education levels (HighSchool, Bachelor, Postgraduate). The analyst includes all three dummies in the regression and also includes an intercept. The regression software returns a singular matrix warning and the coefficient estimates are unstable. What is the cause and correct remedy?

- A. The dataset is too small; more observations are needed.
- B. The dummies sum to one for every observation, so together with the intercept they create perfect multicollinearity (the dummy variable trap). Remedy: drop one of the dummies (the baseline category) or remove the intercept.
- C. The dummies should have been coded as 0, 1, 2 for the three education levels.
- D. The dummies should have been normalized before inclusion.

Answer: B

## Explanation

With  $k$  categories and  $k$  dummies, the dummies sum to one identically. Adding an intercept then creates perfect collinearity. The standard remedy is to drop one dummy (the baseline) or remove the intercept. Option A is wrong because sample size is not the issue. Option C is wrong because ordinal encoding of nominal categories is inappropriate. Option D is wrong because scaling dummy variables does not fix the collinearity.

## Question

A risk analyst examines the distribution of a feature representing “individual trade size in USD”. The minimum is \$100 and the maximum is \$14 million. The ratio of max to min is approximately 140,000. A histogram shows a very long right tail. The analyst decides to log-transform the feature. Which of the following best describes the effect of this decision?

- A. The log transformation eliminates outliers from the dataset.
- B. The log transformation typically reduces positive skewness and stabilises the variance of the feature, making it more suitable as an input to linear models that assume well-behaved residuals.
- C. The log transformation converts the feature into a categorical variable.
- D. The log transformation has no effect on the shape of the distribution; only scaling methods change the shape.

Answer: B

## Explanation

Log transformations reduce right skewness and stabilise variance in features with multiplicative structure (common in financial data like trade sizes, incomes, or market caps). Option A is wrong — logs do not delete data points. Option C is wrong — logs keep the feature numeric. Option D is wrong because log transformations do change the shape, unlike scaling.

## Question

A fixed-income analyst applies PCA to daily changes in eight US Treasury yields (1Y, 2Y, 3Y, 5Y, 7Y, 10Y, 20Y, 30Y). The first principal component explains 82% of the variance and its loadings are all positive and approximately equal. The second PC explains 8% with loadings negative on short maturities and positive on long maturities. The third PC explains 5% with loadings positive on the very short end, negative in the middle, and positive on the long end. How should these three PCs be interpreted?

- A. PC1 = level (parallel shift); PC2 = slope (steepening/flattening); PC3 = curvature (butterfly).
- B. PC1 = slope; PC2 = level; PC3 = curvature.
- C. PC1 = curvature; PC2 = level; PC3 = slope.
- D. All three PCs are uninterpretable and should be discarded.

Answer: A

## Explanation

The first PC has equal positive loadings across all maturities, representing a parallel shift (level). The second PC has opposite signs on the short versus long ends, representing a change in slope. The third PC has signs that go positive, negative, positive across maturities, representing curvature (butterfly). This is a classic result and widely used in fixed-income risk management.

## Question

A risk team is comparing three candidate models to predict the severity of operational losses: (i) ordinary least squares regression with a linear specification, (ii) a decision tree regressor, and (iii) a support vector machine with an RBF kernel. The dataset contains several extreme losses caused by one-off events. Which statement best captures the relative robustness of these three models to outliers?

- A. All three models are equally sensitive to outliers because they all minimize squared error.
- B. OLS is the most sensitive because squared residuals heavily penalise large deviations; the tree and the SVM are typically more robust because they do not rely on a globally quadratic loss.
- C. The tree is the most sensitive because it splits on thresholds, which are distorted by extreme values.
- D. The SVM is the most sensitive because the RBF kernel amplifies distances.

Answer: B

## Explanation

OLS uses a squared loss that heavily penalises large residuals, so a single extreme observation can substantially distort the fitted line. Decision trees split on thresholds and are less sensitive to the exact magnitude of a single point. SVMs with margins are also relatively robust. Option A is wrong because the loss functions differ. Option C is wrong because trees are typically more robust to outliers, not less. Option D is wrong because the RBF kernel is a modelling choice; the SVM's robustness comes from its margin formulation.

## Question

A team building a credit-default classifier has 1,000 labelled observations with a 95/5 class imbalance. They consider three evaluation schemes: (i) a single 70/15/15 train/validation/test split, (ii) 10-fold cross-validation, (iii) 10-fold *stratified* cross-validation. Which scheme is most appropriate, and why?

- A. Scheme (i), because a single split avoids leakage.
- B. Scheme (ii), because k-fold uses all data.
- C. Scheme (iii), because stratification ensures each fold preserves the 95/5 class ratio, avoiding folds with zero minority-class examples and yielding a less biased performance estimate.
- D. None of the schemes are appropriate; oversampling should be applied first.

Answer: C

## Explanation

With severe class imbalance, standard k-fold may leave some folds without minority examples, biasing the estimate. Stratified k-fold preserves the class ratio in every fold. Option A is wrong because a single split on 1,000 observations with 5

## Question

An AML alert system has precision 0.35 and recall 0.85. The head of compliance asks what this means and whether the model is appropriate. Which of the following responses is most accurate?

- A. Precision 0.35 means 35% of actual money-laundering events are caught; recall 0.85 means 85% of alerts are real. The model is excellent.
- B. Precision 0.35 means that 35% of alerts are real suspicious activity; recall 0.85 means that 85% of actual suspicious activity is caught. The model is tuned to favour recall, which is often appropriate for AML given the cost of missing true positives.
- C. Precision 0.35 means 35% accuracy; recall 0.85 means 85% accuracy on positive cases.
- D. Precision and recall are equivalent metrics and their values should always be equal.

Answer: B

## Explanation

Precision =  $TP/(TP+FP)$  = proportion of alerts that are real positives. Recall =  $TP/(TP+FN)$  = proportion of actual positives caught. In AML, false negatives (missed suspicious activity) are far costlier than false positives (extra investigations), so a high-recall, moderate-precision model is often desirable. Option A reverses the definitions. Option C confuses with accuracy. Option D is wrong.

## Question

A fraud-detection model has accuracy 99.4% and AUC 0.62. A second model has accuracy 97.1% and AUC 0.91. The fraud rate is 0.3%. The head of risk asks which model is better and why.

- A. Model 1 is better because higher accuracy is the ultimate criterion.
- B. Model 2 is better because, in a highly imbalanced setting, accuracy is misleading; AUC measures the model's ability to discriminate between classes across all thresholds, which is the relevant property.
- C. Both models are equivalent because accuracy and AUC are two views of the same quantity.
- D. Neither model is usable because fraud data is always too imbalanced to model.

Answer: B

## Explanation

When the positive class is rare (here 0.3%), accuracy is a poor metric — a model that predicts all negatives will achieve 99.7% accuracy but be useless. AUC measures discrimination and is robust to class imbalance. Model 2 is clearly better despite lower accuracy. Option A ignores the imbalance. Option C is wrong because accuracy and AUC measure different properties. Option D is a false generalisation.

## Question

A risk manager must choose a classification threshold for a credit approval model. The cost of approving a defaulting borrower (false positive) is estimated at \$50,000. The cost of rejecting a good borrower (false negative) is estimated at \$2,000 in foregone profit. The current threshold of 0.5 produces far more false positives than false negatives. Which direction should the risk manager move the threshold, and why?

- A. Lower the threshold, because the model should approve more borrowers.
- B. Raise the threshold (require a higher probability of default to reject), because the ratio of costs implies that the model should be more willing to approve marginal cases and avoid the much smaller cost of false negatives.
- C. Keep the threshold at 0.5, because it is the theoretically optimal choice in all cases.
- D. Raise the threshold, because false negatives are always more expensive than false positives.

Answer: B

## Explanation

The threshold should be set by the cost matrix. Here, the cost of a false positive (approving a defaulter, \$50,000) is far greater than the cost of a false negative (rejecting a good borrower, \$2,000). Wait — this reverses the direction. If FP is much more costly, the model should be more conservative: raise the threshold so that more marginal cases are rejected (fewer FPs), accepting more FNs. So the correct action is to *raise* the threshold when FP cost exceeds FN cost. Option B incorrectly says lower; Option B's justification is inverted. The correct answer under the stated cost asymmetry is D — but only if we recognise that "raise the threshold" means "require a higher PD to reject". Actually: raising the threshold means fewer rejections (fewer positives predicted), so fewer FPs. Given FP cost is  $25\times$  FN cost, raising the threshold is correct. Option D's reasoning is right but generic; B's reasoning is wrong.

## Question

A data scientist is building a predictive model for corporate bond defaults with 300 candidate features, of which perhaps 20 are truly informative. The features include several groups of highly correlated financial ratios. The analyst is deciding between Ridge, LASSO, and Elastic Net. Which choice is best, and why?

- A. Ridge, because it shrinks all coefficients smoothly and never eliminates features.
- B. LASSO, because it eliminates irrelevant features entirely, producing a sparse model.
- C. Elastic Net, because it combines the sparse-selection property of LASSO with Ridge's ability to handle groups of highly correlated features.
- D. No regularization, because regularization is only for neural networks.

Answer: C

## Explanation

With 300 features and correlated groups, LASSO alone may arbitrarily select one feature from each correlated group and drop the others. Ridge shrinks but keeps all features. Elastic Net combines L1 sparsity with L2 grouping — the best of both worlds when there are correlated groups of features. Option A forgoes sparsity. Option B is vulnerable to arbitrary selection. Option D is categorically wrong.

## Question

A junior analyst builds a predictive model and proudly reports 99.8% accuracy on the test set. The senior data scientist is suspicious and asks to review the code. She notices that feature scaling was applied to the full dataset before splitting into train and test. She also notices that a target-encoding feature was computed using the full dataset. Which problem has most likely occurred?

- A. Overfitting to noise, because too many features were used.
- B. Data leakage, because both the scaling parameters and the target-encoded feature were computed using test-set information and then exposed to the training process.
- C. Underfitting, because the model is too simple.
- D. Nothing is wrong; these preprocessing steps are standard.

Answer: B

## Explanation

Both scaling parameters (mean, SD) and target encodings must be computed using only the training data. If they are computed over the full dataset before splitting, information from the test set leaks into the training process, producing overly optimistic performance. Option A is not the primary issue here. Option C is the opposite. Option D is wrong — the practice is not standard; it is a known anti-pattern.

## Question

A credit-scoring dataset has a field for “years of employment”. It is missing for 30% of records. Investigating, the analyst finds that most missing values occur in self-employed applicants. The analyst must decide whether to impute, remove rows, or treat missingness as informative. Which of the following is the best description of what is happening and the appropriate response?

- A. The missingness is random; standard mean imputation is sufficient.
- B. The missingness is structural and informative — it correlates with an important applicant characteristic (self-employment) that is itself predictive of credit risk. The analyst should consider adding an “is self-employed” dummy or modelling missingness explicitly rather than imputing.
- C. The missingness is an error and should be treated as zero.
- D. Removing all rows with missing data would be optimal regardless of the mechanism.

Answer: B

## Explanation

Structural missingness correlated with an informative variable is not random. Ignoring it can bias the model. Better approaches include encoding a dummy for “missing” or explicitly modelling self-employment. Option A is wrong because the mechanism is not random. Option C is wrong because zero is a false value. Option D discards valuable data.

## Question

A risk analyst plots a normal Q-Q plot of daily returns on a large equity index. In the tails, the points sit well below the diagonal line on the left and well above the diagonal on the right. The middle of the plot hugs the diagonal. What does this indicate?

- A. The distribution is well-approximated by a normal distribution.
- B. The distribution has fatter tails than the normal distribution on both sides, indicating greater probability of extreme positive and negative returns than the normal model implies.
- C. The distribution is uniform.
- D. The dataset has been mis-scaled.

Answer: B

## Explanation

Deviation from the diagonal at the tails — with points on the extremes pulled away — indicates fat tails (leptokurtosis). This is common in financial return data and has material implications for risk management (e.g., VaR underestimation under normality assumptions). Options A, C, and D are incorrect.

## Question

An analyst is building a churn model with a continuous feature “monthly spending in USD”. A senior colleague suggests discretising this feature into quartile bins, arguing that this makes the model more robust to outliers and easier to interpret. Another colleague argues against discretization because it discards information. What is the best trade-off consideration?

- A. Discretization is always better because it simplifies the model.
- B. Discretization is always worse because it destroys information.
- C. Discretization can improve robustness and interpretability, but it also discards within-bin variance and can introduce arbitrary boundaries; for linear models, keeping the continuous feature is often preferable unless there is a clear non-linear pattern or a regulatory reason to bin.
- D. Discretization should only be applied to categorical features.

Answer: C

## Explanation

Discretization is a trade-off: it improves interpretability and robustness to noise but sacrifices information and can introduce arbitrary bin boundaries. Whether to use it depends on the modelling context (linear vs tree-based, regulatory interpretability requirements, etc.). Options A and B are both one-sided. Option D misstates what discretization is applied to.

## Question

A team is developing a machine learning model on 500,000 labelled credit events. The team lead proposes to allocate 90% to training, 5% to validation, and 5% to test. An alternative proposal is 60/20/20. Which is more defensible and why?

- A. 90/5/5 is better because training on more data improves model performance.
- B. 60/20/20 is better because validation and test samples of 100,000 events each give much more reliable model-selection and performance-estimation results, while 450,000 training samples is already well beyond the point of diminishing returns.
- C. 90/5/5 is better because the test set only needs to be large enough for a rough estimate.
- D. Both are equally appropriate; the split ratio has no effect on results.

Answer: B

## Explanation

With 500,000 events, training on 450,000 or 300,000 is unlikely to make much difference; both are large enough for most models. But validation and test sets of 25,000 vs 100,000 events make a large difference in the reliability of model selection and performance estimation. Option B captures this trade-off. Option A is a common but naive argument. Option D is wrong.

## Question

A correlation matrix of 10 features in a credit model shows that two features have a pairwise correlation of 0.97. Their individual correlations with the target variable are 0.42 and 0.41. What is the most likely consequence of including both in an ordinary least squares regression?

- A. Both coefficients will be estimated accurately and their t-statistics will be large.
- B. The regression will fail to converge.
- C. The coefficient estimates will be unstable with inflated standard errors, and their signs may reverse relative to economic intuition.
- D. The coefficients will be identical to each other.

Answer: C

## Explanation

High pairwise correlation between two predictors (0.97) causes multicollinearity. The OLS estimators remain unbiased but become unstable, with large standard errors and frequently counterintuitive signs. Remedies include dropping one variable, combining them (e.g., averaging), or using PCA or regularisation. Option A is the opposite of what occurs. Option B is wrong for near-collinearity (only perfect collinearity fails). Option D is not guaranteed.

## Question

A quantitative researcher must choose between stepwise regression, best subset selection, and LASSO for feature selection in a model with 40 candidate predictors. Which of the following statements is most accurate?

- A. Best subset evaluates all  $2^{40}$  subsets and is therefore practical for any modern computer.
- B. Stepwise regression finds the true optimal subset but is slower than LASSO.
- C. Best subset is theoretically optimal but computationally infeasible for 40 features; stepwise is a fast heuristic that may miss the optimal subset; LASSO provides a convex alternative that combines selection with shrinkage, though it can be unstable when features are highly correlated.
- D. LASSO is always superior to both stepwise and best subset.

Answer: C

## Explanation

Best subset is exponential in the number of features and impractical beyond 30–40 features. Stepwise is a greedy heuristic that may miss the optimum. LASSO is convex and scalable but unstable with correlated features. This balanced view is correct. Options A, B, and D are either wrong or oversimplifications.

## Question

PCA on Treasury yields produces a first component explaining 82% of variance, a second explaining 8%, a third explaining 4%, and the remaining five explaining 6% combined. A portfolio manager wants a compact factor model. How many PCs should the manager retain?

- A. All eight, because PCA is designed to be used with all components.
- B. One, because it dominates the variance decomposition.
- C. Three, because they together explain roughly 94% of variance and correspond to interpretable level/slope/curvature factors, while the remaining five are largely idiosyncratic noise.
- D. Five, because they explain at least 99% of variance.

Answer: C

## Explanation

A scree plot or cumulative-variance analysis typically shows an elbow after the first 2–3 PCs in yield-curve data. Retaining three PCs captures 94% of variance and yields interpretable factors (level, slope, curvature). Options A and B are extremes. Option D is factually wrong — the sum here is  $82+8+4+6 = 100\%$ , so five components would be four extra. In practice, PC1–PC3 is the standard for a compact model.

## Question

A data scientist applies min-max normalization to every feature. A colleague claims that normalization “destroys the correlation structure of the data”. Which statement is correct?

- A. The colleague is correct — normalization changes the correlation matrix.
- B. The colleague is incorrect — min-max normalization is an affine transformation of each feature individually and preserves Pearson correlations between features.
- C. The colleague is correct only if the features are skewed.
- D. Correlation is not affected by any scaling technique; the claim is unfounded either way.

Answer: B

## Explanation

Min-max normalization is a linear transformation applied independently to each feature. Pearson correlations between features are invariant under per-feature affine transformations. Option A is wrong. Option C is wrong — skewness does not change this property. Option D is too broad; some transformations (e.g., non-monotone ones) can change correlations.

## Question

A data platform team must classify incoming datasets for storage and processing. Which of the following is best classified as semi-structured data?

- A. A CSV file with a fixed schema of 10 columns.
- B. An email message with header fields (From, To, Date, Subject) and a free-text body.
- C. A JPEG photograph.
- D. A relational database table.

Answer: B

## Explanation

Emails have a defined header structure (metadata) but free-form body — the classic example of semi-structured data. CSV and relational tables are structured. JPEGs are unstructured. Option A is structured; Option C is unstructured; Option D is structured.

## Question

A data scientist wants to encode a high-cardinality categorical variable "region" (with 500 levels) using target encoding, which replaces each level with the average target value in that level. She computes the encodings on the full dataset (train + test) before splitting. What is the consequence?

- A. No consequence; target encoding is always safe.
- B. Data leakage — the test set's target information has been used to construct the training encodings, producing optimistically biased performance estimates.
- C. The model will underfit.
- D. The categorical variable should have been one-hot encoded instead, which is always superior.

Answer: B

## Explanation

Target encoding must be computed using only the training data, and ideally within each cross-validation fold to avoid leakage. Computing on the full dataset leaks test-set target information. Option A is wrong — target encoding is notoriously leak-prone. Option C is the opposite. Option D is not universally true; one-hot is a different approach with different trade-offs.

## Question

In a linear regression of loan severity on borrower characteristics, one observation has a Cook's distance 20 times larger than the median. What does this indicate, and what should the analyst do?

- A. The observation has a large residual and is highly influential on the fitted coefficients. The analyst should investigate whether the observation is a data-entry error, a genuine outlier, or a case that warrants special treatment.
- B. The observation is a measurement error and should be removed automatically.
- C. Cook's distance is unrelated to influence and can be ignored.
- D. The observation is a normal point and should be retained without investigation.

Answer: A

## Explanation

Cook's distance measures the influence of each observation on the fitted regression coefficients. A large value indicates high influence. The correct response is investigation — not automatic deletion. Option B is too aggressive. Option C is wrong. Option D ignores a real signal.

## Question

An analyst argues that feature scaling should be applied to the entire dataset before splitting into train and test, so that the mean and standard deviation are “more representative”. Another analyst disagrees. Who is correct and why?

- A. The first analyst is correct — the entire dataset should be used to compute reliable scaling parameters.
- B. The second analyst is correct — scaling parameters must be computed on the training set and then applied to the test set, to avoid data leakage and ensure the test set remains a fair out-of-sample evaluation.
- C. Both approaches are equivalent.
- D. Neither is correct; scaling should not be used at all.

Answer: B

## Explanation

Scaling parameters (mean, SD) must be estimated from the training data only, then applied to the test data, to simulate the real deployment scenario where future data is scaled using the training distribution. Using the full dataset leaks test information and inflates performance. Option A is a common mistake. Option C is wrong. Option D is wrong — scaling is often essential.

## Question

A quantitative researcher argues that deep neural networks can approximate any continuous function given sufficient depth and width. A risk manager asks whether this means a single very large deep network is sufficient for all risk modelling tasks. Which response is most accurate?

- A. Yes — the universal approximation theorem guarantees that a deep network can model any risk phenomenon without feature engineering or regularisation.
- B. No — the theorem guarantees representational capacity but says nothing about learnability from finite data, computational cost, or generalisation; in practice, careful feature engineering, architecture choice, and regularisation remain essential.
- C. Yes, but only if the network has exactly three hidden layers.
- D. No, because the theorem only applies to linear functions.

Answer: B

## Explanation

The universal approximation theorem states that certain neural networks can approximate any continuous function on a compact domain to arbitrary accuracy — but it does not guarantee that such a network can be *learned* from finite data within reasonable resources, nor that it will generalise. Options A, C, and D misstate the theorem.

## Question

An analyst is fitting both Ridge and LASSO regressions on a set of features with vastly different scales (e.g., age in years vs income in dollars). Why is standardizing features before regularization considered best practice?

- A. It is only important for LASSO, not for Ridge.
- B. The penalty term in Ridge or LASSO is applied to coefficient magnitudes; if features are on different scales, a coefficient on a large-scale feature can be small in magnitude but still have a large effect, so the penalty would be unfairly applied. Standardizing ensures penalties are applied uniformly.
- C. Standardization removes outliers.
- D. Standardization guarantees the model will have zero bias.

Answer: B

## Explanation

Regularization penalties operate on coefficient magnitudes. If a feature is on a large scale, its coefficient can be small yet impactful, and the penalty would under-shrink it relative to a small-scale feature. Standardizing features puts all coefficients on comparable footing. Option A is wrong — both Ridge and LASSO benefit. Options C and D are incorrect claims.

## Question

A model has very low training error and very high test error. Which quadrant of the bias-variance decomposition does this correspond to, and what remedy is appropriate?

- A. High bias, low variance — remedy: increase model complexity.
- B. Low bias, high variance — remedy: add regularization, simplify the model, or gather more data.
- C. Low bias, low variance — remedy: no change needed.
- D. High bias, high variance — remedy: reduce feature count and increase model complexity simultaneously.

Answer: B

## Explanation

Low training error + high test error is the signature of overfitting: low bias, high variance. Remedies include regularization (L1/L2), reducing model complexity, gathering more data, or using ensembling. Option A is the opposite quadrant. Option C is ideal and does not fit. Option D describes a different (unusual) scenario.

## Question

A team is considering two approaches to estimate the relationship between firm size and default probability: a logistic regression with a linear index, and a KNN classifier with  $k=5$ . Which statement best captures the trade-offs?

- A. KNN is a parametric model and logistic regression is non-parametric.
- B. Logistic regression is parametric and imposes a specific functional form; KNN is non-parametric and makes no functional-form assumption but requires more data to achieve comparable accuracy and can suffer from the curse of dimensionality.
- C. Both are non-parametric.
- D. Logistic regression is always preferable because it is interpretable.

Answer: B

## Explanation

Parametric models impose a functional form (e.g., linear-in-log-odds for logistic regression). Non-parametric models (KNN, decision trees, kernel methods) are flexible but data-hungry and prone to the curse of dimensionality. Option A reverses the categories. Option C is wrong. Option D is too strong a claim — interpretability is a trade-off, not a universal criterion.

## Question

A team building a forecasting model for quarterly revenues has 40 quarters of data. A junior analyst proposes standard 5-fold cross-validation. Why is this problematic and what should be used instead?

- A. Standard k-fold shuffles the data, allowing future quarters to appear in training while earlier quarters are in validation — this produces an inflated performance estimate. Time-series cross-validation (e.g., rolling or expanding window) should be used instead.
- B. Standard k-fold is fine for any dataset size.
- C. The 5-fold CV is a problem only because the sample is small.
- D. Time-series data requires no validation at all.

Answer: A

## Explanation

Standard k-fold assumes observations are i.i.d. and can be randomly assigned. For time series, this violates the temporal ordering and causes leakage. Rolling- or expanding-window CV respects temporal order. Option B ignores the issue. Option C misses the real reason. Option D is wrong.

## Question

A risk team fits a logistic regression with 200 features to predict defaults on 500 observations. The unregularized model perfectly separates the training data (all predictions correct), but test performance is poor. Why did this happen, and what would you recommend?

- A. This is normal; the model is simply better on training data.
- B. Perfect training separation with a large feature-to-observation ratio indicates overfitting. Regularized logistic regression (L1 or L2 penalty) is the standard remedy, with hyperparameter tuning via cross-validation.
- C. The solution is to add more features.
- D. The solution is to remove the target variable.

Answer: B

## Explanation

Perfect separation on training data with  $p \gg n$  is a classic overfitting signature. Regularization shrinks coefficients and improves generalisation. Option A is a mistaken reading. Options C and D are nonsensical.

## Question

In a model with a group of highly correlated features (say, five measures of firm profitability), LASSO tends to select one and drop the rest, arbitrarily. Ridge tends to shrink all five without eliminating any. What is the practical implication, and what does Elastic Net offer?

- A. LASSO is always best because it produces the sparsest model.
- B. Ridge is always best because it keeps all features.
- C. LASSO's arbitrary selection among correlated features can harm interpretation and stability; Elastic Net combines L1 (sparsity) with L2 (grouping), often yielding better stability and predictive performance with correlated feature groups.
- D. None of them matter in practice.

Answer: C

## Explanation

When features are correlated, LASSO's selection is unstable — small changes in data can lead to different features being selected. Ridge keeps all features but does not perform selection. Elastic Net combines both strengths. Option C correctly captures this.

## Question

A modeller is fitting a neural network to predict portfolio returns. Unlike linear regression, there is no closed-form solution for the optimal parameters. Which approach is used, and why?

- A. Ordinary least squares is used, because it works for any model.
- B. Nonlinear least squares via numerical iteration (e.g., gradient descent with backpropagation) is used, because the objective function is non-convex in the parameters and no analytic solution exists.
- C. Maximum likelihood estimation cannot be used.
- D. Parameters are set randomly and never updated.

Answer: B

## Explanation

Nonlinear least squares solves the same objective (minimise squared residuals) numerically when closed-form solutions are unavailable. For neural networks, gradient descent with backpropagation is standard. Option A is wrong — OLS only has closed-form solutions for linear-in-parameters models. Option C is wrong — MLE is also used for such models. Option D is wrong.

## Question

Which of the following best describes the difference between OLS and MLE when estimating a linear regression with normally distributed errors?

- A. OLS and MLE produce different coefficient estimates in general.
- B. OLS and MLE produce identical coefficient estimates when errors are i.i.d. normal with constant variance; MLE is a more general framework that extends to non-linear and non-normal models.
- C. MLE is always worse than OLS.
- D. OLS requires knowing the distribution of errors; MLE does not.

Answer: B

## Explanation

Under i.i.d. normal errors, the MLE and OLS estimators of the linear regression coefficients coincide. MLE is more general — it extends to logistic regression, GARCH models, and many others. Option A is wrong. Option C is wrong. Option D reverses the requirement.

## Question

A data scientist is training a deep neural network. She compares batch gradient descent (using all samples per update), stochastic gradient descent (one sample per update), and mini-batch gradient descent (e.g., 32 samples per update). Which statement is most accurate?

- A. Batch GD always converges faster in wall-clock time than SGD.
- B. SGD requires far fewer updates than batch GD to reach the optimum.
- C. Mini-batch GD combines the computational efficiency of vectorized operations with the stochasticity that helps escape poor local minima; it is the practical default for deep learning.
- D. All three variants are mathematically identical.

Answer: C

## Explanation

Mini-batch GD is the industry default because it exploits GPU parallelism and provides enough stochasticity to help optimisation. Batch GD has smooth but slow updates per data pass; SGD is noisy but converges quickly per pass; mini-batch is a practical middle ground. Options A, B, and D are incorrect.

## Question

A deep neural network with 30 sigmoid-activated hidden layers trains extremely slowly and the early layers' weights barely change. What is the most likely cause, and what remedies exist?

- A. Overfitting — remedy: more data.
- B. Vanishing gradient — remedies: use ReLU or similar activations, batch normalization, or architectures like LSTM/ResNet that mitigate gradient decay.
- C. Underfitting — remedy: more features.
- D. The learning rate is too large — remedy: reduce it.

Answer: B

## Explanation

With sigmoid activations and many layers, gradients propagated by the chain rule shrink exponentially, causing early layers to learn very slowly (the vanishing gradient problem). ReLU, batch norm, and residual connections are standard remedies. Options A, C, and D do not address the root cause.

## Question

Why does adding a momentum term to gradient descent usually improve convergence?

- A. It reduces the learning rate automatically.
- B. It accumulates a velocity vector in directions of persistent gradient, dampening oscillations and accelerating progress along narrow valleys of the loss surface.
- C. It eliminates local minima entirely.
- D. It converts the problem into a convex one.

Answer: B

## Explanation

Momentum smooths the optimisation trajectory, accelerating consistent gradient directions and reducing zig-zagging. It does not eliminate local minima and does not change convexity. Options A, C, and D are incorrect.

## Question

A hiring algorithm trained on 10 years of historical hiring decisions shows a bias against female candidates, even though gender is not an input feature. Which mechanism best explains this phenomenon?

- A. Random noise.
- B. Proxy discrimination — features correlated with gender (e.g., university, job titles, career gaps) carry the historical bias even when gender itself is removed.
- C. The dataset is too small.
- D. The model has too many features.

Answer: B

## Explanation

Removing a protected attribute does not remove its influence if correlated proxies remain. This is proxy discrimination, a well-documented phenomenon. Options A, C, and D do not explain it.

**Question**

A production fraud-detection model suddenly starts producing twice as many alerts as usual. The model has not been retrained. Which is the most likely explanation?

- A. The model has been compromised.
- B. Input data drift or concept drift has changed the distribution of features or the relationship between features and target, causing the model's predictions to shift. The appropriate response is to investigate the drift and consider recalibration or retraining.
- C. The alert threshold has been changed.
- D. This is expected and requires no action.

**Answer:** B

**Explanation**

Data drift (feature distribution change) and concept drift (feature-target relationship change) are common causes of model performance degradation in production. Monitoring and retraining are standard responses. Options A, C, and D are unlikely or wrong.

## Question

A data scientist reports results that a colleague cannot reproduce. The model uses random initialisation for a neural network. What is the most likely cause of the reproducibility problem?

- A. The neural network is broken.
- B. Random seeds were not fixed, so each run started from different initial weights and stochastic training paths, producing different final models.
- C. The dataset has been corrupted.
- D. Neural networks are inherently non-reproducible.

Answer: B

## Explanation

Reproducibility requires fixing random seeds for weight initialisation, data shuffling, and dropout (if used). Without this, results vary. Option A is wrong. Option C is unlikely. Option D is a misconception — with fixed seeds, neural network training is deterministic.

## Question

An ensemble of 100 decision trees trained on random subsets of features (random forest) performs much better than an ensemble of 100 trees trained on the same features (bagging). Why?

- A. Random forests use fewer trees.
- B. Feature subsampling decorrelates individual trees, reducing the variance of the ensemble more effectively than bagging alone.
- C. Bagging is not a valid ensemble technique.
- D. Random forests have a closed-form solution.

Answer: B

## Explanation

Ensemble variance depends on both individual model variance and pairwise correlations between models. Random feature subsampling decorrelates the trees, reducing ensemble variance more than bagging alone. Options A, C, and D are incorrect.

## Question

A quantitative researcher compares AdaBoost and random forest on a complex fraud-detection dataset. Which statement is most accurate?

- A. Boosting trains models in parallel; bagging trains them sequentially.
- B. Bagging trains models in parallel and reduces variance; boosting trains models sequentially, each correcting the errors of the previous, and reduces bias. On complex non-linear data, boosting often achieves lower test error but is more sensitive to noisy labels and outliers.
- C. Boosting is always preferable.
- D. Bagging is always preferable.

Answer: B

## Explanation

Bagging (random forest) reduces variance via parallel decorrelated trees. Boosting (AdaBoost, gradient boosting) sequentially reweights or fits residuals, reducing bias. Boosted models often outperform but are more sensitive to label noise. Option A reverses the parallel/sequential distinction. Options C and D are over-general.

## Question

A retail bank clusters its 5,000 customers using K-means on two features: average monthly spend and number of products held. WCSS values are: K=1: 12,500; K=2: 4,400; K=3: 4,150; K=4: 4,100; K=5: 4,080. The business team wants a small number of interpretable segments. What is the most defensible choice of K?

- A. K=1, because it is simplest.
- B. K=2, because the elbow is between K=1 and K=2, and further reductions in WCSS are marginal relative to the added complexity.
- C. K=5, because it minimizes WCSS.
- D. K=10, because more clusters always yield better segmentation.

Answer: B

## Explanation

The elbow is clearly between K=1 and K=2. Beyond K=2, WCSS reductions are small. Choosing K=2 respects parsimony and interpretability. Options C and D overfit. Option A is too coarse for the dataset.

## Question

A data scientist runs K-means 20 times with different random initialisations and records WCSS values of 98, 100, 101, 101, 102, 99, 250, 100, 101, 99, 102, 100, 101, 99, 100, 101, 102, 99, 100, 101. The single value 250 stands out. What does this most likely indicate, and what should the data scientist do?

- A. The 250 result is the global optimum and should be selected.
- B. The 250 result represents convergence to a poor local optimum. The data scientist should select the run with the lowest WCSS, and can use K-means++ initialisation to reduce the likelihood of such poor local optima.
- C. K-means has failed; switch to hierarchical clustering.
- D. WCSS values above 200 are always indicative of an error.

Answer: B

## Explanation

K-means is not guaranteed to find the global minimum; poor random initialisations can lead to poor local optima. K-means++ spreads initial centroids, reducing this risk. Option A is wrong — higher WCSS is worse. Option C is premature. Option D is a false general rule.

## Question

A cluster solution has low WCSS and high BCSS. A second solution has high WCSS and low BCSS. Which solution is preferred, and what do the metrics measure?

- A. The first solution is preferred. WCSS measures cluster compactness (within-cluster sum of squared distances to centroids); BCSS measures separation (between-cluster sum of squared distances to the overall centroid).
- B. The second solution is preferred because higher WCSS means more information retained.
- C. The two solutions are equivalent.
- D. WCSS and BCSS are alternative names for the same quantity.

Answer: A

## Explanation

WCSS = compactness (lower is better); BCSS = separation (higher is better). A good clustering solution has low WCSS and high BCSS. Option B confuses the direction. Option C is wrong. Option D is factually incorrect.

## Question

In hierarchical clustering, the choice between single linkage and complete linkage has material consequences. Which statement best captures the trade-off?

- A. Single linkage defines cluster distance as the minimum pairwise distance (tends to produce elongated clusters and can chain together clusters via bridging points). Complete linkage defines it as the maximum pairwise distance (tends to produce compact, balanced clusters and is more robust to chaining).
- B. Single linkage is always preferable.
- C. Complete linkage is always preferable.
- D. The linkage criterion only affects the visual representation of the dendrogram, not the clustering outcome.

Answer: A

## Explanation

Single linkage can chain clusters via a single pair of close points, producing elongated shapes. Complete linkage uses the farthest points and produces compact clusters. Neither is universally superior — the choice depends on the data and problem. Option D is wrong; linkage changes the results.

## Question

A data scientist runs DBSCAN on a dataset of transaction locations to detect fraud clusters. She sets  $\epsilon = 0.001$  (very small) and  $\text{minPts} = 2$ . The result is many tiny clusters and most points labelled noise. Which adjustment is most appropriate?

- A. Further decrease  $\epsilon$  and  $\text{minPts}$ .
- B. Increase  $\epsilon$  (larger neighbourhood radius) and/or increase  $\text{minPts}$  so that dense regions are recognised and clusters merge appropriately.
- C. Switch to K-means.
- D. Disable noise labelling.

Answer: B

## Explanation

Very small  $\epsilon$  fragments clusters and marks most points as noise. Increasing  $\epsilon$  allows more points into each neighbourhood; adjusting  $\text{minPts}$  controls the density threshold. Option A worsens the problem. Option C may be an alternative but does not address parameter tuning. Option D is not a DBSCAN feature.

## Question

A dataset contains two interleaved crescent-moon shaped clusters. A data scientist applies K-means with  $K=2$  and gets poor results. Why does K-means fail here, and which algorithm is better suited?

- A. K-means fails because it uses Euclidean distance to centroids and therefore assumes roughly spherical, convex clusters. DBSCAN (density-based) or spectral clustering is better suited for non-convex shapes.
- B. K-means fails because  $K$  is too large;  $K=1$  would work.
- C. K-means fails because the data has too few dimensions.
- D. K-means fails because it cannot handle non-numeric data.

Answer: A

## Explanation

K-means's centroid-based objective favours spherical, similar-sized clusters. For crescent-shaped clusters, density-based methods (DBSCAN) or graph-based methods (spectral clustering) are appropriate. Options B, C, and D are wrong.

## Question

A clustering with  $K=2$  has average silhouette score 0.71. With  $K=3$ , the average silhouette is 0.45. With  $K=4$ , it is 0.62. Which  $K$  is preferred based on silhouette analysis?

- A.  $K=2$ , because it has the highest silhouette score.
- B.  $K=3$ , because more clusters are better.
- C.  $K=4$ , because it is close to  $K=2$ .
- D. Cannot be determined from silhouette scores alone.

Answer: A

## Explanation

Silhouette scores range from -1 to 1, with higher values indicating better-separated, more cohesive clusters.  $K=2$  has the highest score and is preferred. Options B and C ignore the numeric criterion. Option D is wrong — the criterion is explicitly designed for this decision.

## Question

A CNN accepts an input of size  $32 \times 32 \times 3$  (RGB image). A convolutional layer uses  $3 \times 3$  kernels, stride 1, and no padding, and outputs 16 channels. What is the shape of the output tensor?

- A.  $32 \times 32 \times 16$ .
- B.  $30 \times 30 \times 16$ .
- C.  $30 \times 30 \times 3$ .
- D.  $16 \times 16 \times 3$ .

Answer: B

## Explanation

Spatial output size =  $(n - k + 1) = 32 - 3 + 1 = 30$ . Channel depth is set by the number of filters = 16. So the output is  $30 \times 30 \times 16$ . Option A ignores the effect of kernel size. Option C keeps the input channels. Option D misapplies the formula.

## Question

An equity research team is building a direction-of-return classifier. They consider random forest and XGBoost. Which statement best captures the practical trade-offs?

- A. Random forest is unbiased and XGBoost is biased.
- B. Random forest trains trees independently (parallelisable, robust to hyperparameters); XGBoost trains trees sequentially, fits residuals, and often achieves higher accuracy at the cost of more careful tuning and greater sensitivity to noisy labels.
- C. XGBoost is a type of random forest.
- D. Both are forms of logistic regression.

Answer: B

## Explanation

Random forest = bagged decorrelated trees; parallel; robust. XGBoost = gradient boosting; sequential; higher accuracy on many tabular tasks but more tuning. Option A is nonsense. Option C is wrong — they are distinct. Option D confuses tree ensembles with regression.

## Question

A risk analyst fits an SVM to a binary classification problem. The linear kernel underperforms; the RBF kernel performs much better. What does this indicate?

- A. The classes are linearly separable.
- B. The decision boundary is non-linear, and the RBF kernel implicitly maps inputs to a higher-dimensional space where a linear separator exists.
- C. The RBF kernel is always superior.
- D. The data is noisy.

Answer: B

## Explanation

The RBF kernel allows the SVM to construct non-linear decision boundaries via the kernel trick. When linear kernels underperform, the classes are typically non-linearly separable. Option A contradicts the observed result. Option C is too broad. Option D is unsupported by the information given.

## Question

A fraud-detection model has accuracy 99.8%, precision 0.05, and recall 0.80. Which statement correctly interprets these values?

- A. The model is excellent because accuracy is very high.
- B. Accuracy is misleading due to class imbalance. The low precision (5%) means most alerts are false positives, but the high recall (80%) means most true frauds are caught. The model might still be useful if investigative costs are acceptable, but threshold tuning or class rebalancing is warranted.
- C. The model is useless because precision is low.
- D. Precision and recall should always sum to 1.

Answer: B

## Explanation

With rare positives, accuracy is a poor metric. Precision =  $TP/(TP+FP)$ ; recall =  $TP/(TP+FN)$ . Low precision means many false alarms; high recall means most fraud is caught. Whether the model is usable depends on cost-benefit considerations. Option A ignores precision. Option C is too strong. Option D is wrong.

## Question

Model A has AUC 0.92 on a test set; Model B has AUC 0.78. Both are logistic regressions on the same features, differing only in regularisation strength. The team must choose one model for production. Which is correct?

- A. Model A, because higher AUC indicates better discrimination across all classification thresholds.
- B. Model B, because lower AUC means the model is more conservative.
- C. Both are equivalent.
- D. AUC is not a valid metric for logistic regression.

Answer: A

## Explanation

AUC measures the model's ability to discriminate between classes across all thresholds. Higher AUC generally indicates a better model, though business considerations (threshold choice, cost matrix) still matter. Options B, C, and D are wrong.

## Question

Two models classify spam emails. Model 1 has precision = 0.95, recall = 0.30. Model 2 has precision = 0.70, recall = 0.85. Which model has the higher F1 score, and what does this mean?

- A. Model 1, because precision is more important.
- B. Model 2, because  $F1 = 2PR/(P+R)$ : Model 1  $F1 = 0.456$ ; Model 2  $F1 = 0.767$ . F1 balances precision and recall, favouring Model 2 in this case.
- C. Equal F1.
- D. Cannot compute.

Answer: B

## Explanation

F1 is the harmonic mean of precision and recall:  $F1 = 2PR/(P+R)$ . Model 1:  $2(0.95)(0.30)/(1.25) = 0.456$ . Model 2:  $2(0.70)(0.85)/(1.55) = 0.767$ . Model 2 wins because it balances precision and recall better.

## Question

An SVM has learned the decision function  $f(x) = 2.5x_1 - 1.8x_2 + 0.4$ . A new observation has  $(x_1 = 1.0, x_2 = 1.5)$ . What is the predicted class, and why?

- A. Class +1, because  $f(x) > 0$ .
- B. Class -1, because  $f(x) < 0$ .
- C. Cannot determine without the kernel.
- D. The decision value is 0, so the observation is on the boundary.

Answer: A

## Explanation

$f(1.0, 1.5) = 2.5(1.0) - 1.8(1.5) + 0.4 = 2.5 - 2.7 + 0.4 = 0.2 > 0$ , so the prediction is the positive class. Option B miscalculates. Option C is unnecessary — the kernel affects the geometry of the boundary but the decision function value determines the class. Option D is wrong because 0.2 not equal 0.

## Question

A fraud analyst compares KNN with Euclidean distance and KNN with cosine distance on a high-dimensional text-feature dataset. The cosine-distance version performs much better. Why?

- A. Euclidean distance is always wrong.
- B. In high-dimensional sparse feature spaces (typical of text), Euclidean distance is dominated by vector magnitude differences rather than direction, whereas cosine distance measures angular similarity and captures semantic direction. This makes cosine distance more effective.
- C. Cosine distance is easier to compute.
- D. KNN cannot use Euclidean distance in high dimensions.

Answer: B

## Explanation

In high-dimensional text data, cosine similarity (angle) is more meaningful than Euclidean distance (which conflates magnitude with direction). This is why cosine distance is standard in NLP. Option A is too strong. Option C is not the reason. Option D is wrong.

## Question

A document classifier uses Naive Bayes on bag-of-words features. The algorithm assumes conditional independence of words given the class. In practice, words in natural language are highly dependent. Why does Naive Bayes often work well anyway?

- A. The independence assumption is always satisfied in practice.
- B. Even with strong violations, Naive Bayes often produces correct *ranking* of class probabilities, and classification decisions depend only on which class has the highest posterior — not on exact probability calibration.
- C. Naive Bayes is equivalent to logistic regression.
- D. The assumption does not matter because the model is unsupervised.

Answer: B

## Explanation

Naive Bayes is known to work well even when the independence assumption is violated because classification depends on the  $\text{argmax}$  of the posterior — ranking, not exact calibration, is what matters. Options A, C, and D are incorrect.

## Question

A research team fits LASSO to a dataset with 500 predictors. Of these, 300 coefficients are estimated as exactly zero. Which statement best describes what happened?

- A. The zeros are numerical errors.
- B. The L1 penalty in LASSO produces sparse solutions by driving irrelevant coefficients to exactly zero, effectively performing automatic feature selection.
- C. Ridge regression would have produced the same result.
- D. The zeros indicate that the model has not converged.

Answer: B

## Explanation

LASSO's L1 penalty has a geometric property that leads to exact zeros, giving sparsity. Ridge (L2) shrinks but rarely to zero. Option A is wrong. Option C is wrong for the reason above. Option D is wrong; convergence of LASSO naturally yields exact zeros.

## Question

A company has 1,000 labelled customer-support transcripts and 100,000 unlabelled ones. It considers two approaches: (i) supervised learning using only the 1,000 labels, and (ii) semi-supervised learning using both. Which statement best captures the practical trade-off?

- A. Supervised learning is always better because labels are more reliable.
- B. Semi-supervised learning can substantially improve performance when unlabelled data can be exploited to infer structure, but its success depends on assumptions (smoothness, cluster, manifold) being reasonably satisfied and can degrade if they are violated.
- C. Semi-supervised learning is always superior.
- D. Both approaches are identical.

Answer: B

## Explanation

Semi-supervised learning leverages unlabelled data via assumptions about the geometry of the input distribution. When assumptions hold, it improves performance; when they are violated, it can be worse than supervised-only. Options A, C, and D are too categorical.

## Question

A text-classification task has two natural views of each article: its title words and its body words. An analyst proposes co-training. What is the core idea?

- A. Train two models, one on each view, and have each model label confident unlabelled examples for the other, expanding the labelled training set iteratively.
- B. Train one model on both views.
- C. Use a single model with both views as separate feature sets.
- D. Apply reinforcement learning to labels.

Answer: A

## Explanation

Co-training leverages multiple feature views by training separate models on each view and having them label data for each other. This exploits conditional independence of views. Options B, C, and D miss the point.

## Question

A semi-supervised technique trains a model on labelled data, predicts labels for unlabelled data, adds the most confident predictions to the training set, and retrains. What is this technique called?

- A. Co-training.
- B. Self-training.
- C. Reinforcement learning.
- D. Transfer learning.

Answer: B

## Explanation

Self-training iteratively expands the labelled set with confident pseudo-labels. It is a simple and widely used semi-supervised method. Options A, C, and D refer to different paradigms.

## Question

A trading algorithm learns to buy or sell based on a reward function that penalises drawdowns. It updates its policy based on observed rewards. Which learning paradigm does this represent?

- A. Supervised learning.
- B. Unsupervised learning.
- C. Reinforcement learning.
- D. Semi-supervised learning.

Answer: C

## Explanation

Reinforcement learning learns a policy by interacting with an environment and receiving rewards. Supervised learning uses labelled examples; unsupervised learns structure; semi-supervised combines both. Options A, B, and D are incorrect.

## Question

An algorithmic trading desk must repeatedly choose among  $K$  strategies, learning which is best over time. Early choices are exploratory; later choices should exploit the best-performing strategy. Which RL formulation is most relevant?

- A. Supervised classification.
- B. The multi-arm bandit problem.
- C. Dimensionality reduction.
- D. Regression.

Answer: B

## Explanation

The multi-arm bandit framework formalises the explore-exploit trade-off in repeated decision problems with uncertain rewards. Options A, C, and D are unrelated.

## Question

In reinforcement learning, Monte Carlo methods update value estimates after each complete episode, while temporal difference (TD) methods update after each step. Which is a key advantage of TD?

- A. TD cannot handle continuous state spaces.
- B. TD can learn online, without waiting for episode termination, and often converges faster on problems with long episodes.
- C. TD requires full knowledge of the environment.
- D. Monte Carlo is always superior.

Answer: B

## Explanation

TD methods learn bootstrapped estimates step-by-step, enabling online learning and quicker updates. Monte Carlo requires waiting until the episode ends. Options A, C, and D are wrong.

## Question

A deep neural network trains with very unstable gradients that occasionally blow up to enormous magnitudes. What is this phenomenon called, and what is a common remedy?

- A. Vanishing gradient — remedy: use sigmoid activations.
- B. Exploding gradient — remedy: gradient clipping, careful weight initialisation, and architectures like LSTM.
- C. Overfitting — remedy: more data.
- D. Underfitting — remedy: more features.

Answer: B

## Explanation

Exploding gradients cause unstable training; gradient clipping caps updates. LSTM and residual architectures help. Option A is the opposite problem. Options C and D are unrelated.

## Question

Which of the following best describes the key architectural feature of a recurrent neural network (RNN)?

- A. It processes the entire input in one forward pass without memory.
- B. It maintains a hidden state that is updated at each time step, allowing it to capture sequential dependencies.
- C. It uses convolutional filters on the input.
- D. It applies attention to all tokens simultaneously.

Answer: B

## Explanation

RNNs pass hidden state across time steps, enabling sequence modelling. Options A and C describe feedforward and convolutional networks. Option D describes transformers.

## Question

A transformer model processes an entire sequence at once and computes pairwise attention between all tokens. Which of the following is a direct advantage over RNNs?

- A. Transformers require less memory.
- B. Transformers capture long-range dependencies efficiently and can be parallelised over sequence positions; RNNs suffer from sequential computation and vanishing gradients over long horizons.
- C. Transformers use less training data.
- D. Transformers avoid the need for embeddings.

Answer: B

## Explanation

The attention mechanism allows direct modelling of long-range dependencies and parallel computation across tokens. Options A and C are wrong in general. Option D is wrong — transformers still use embeddings.

## Question

Which of the following correctly distinguishes the CBoW and skip-gram architectures of Word2Vec?

- A. CBoW predicts context from target; skip-gram predicts target from context.
- B. CBoW predicts a target word from its surrounding context (many inputs, one output); skip-gram predicts surrounding context from the target word (one input, many outputs).
- C. Both architectures are identical.
- D. CBoW is supervised; skip-gram is unsupervised.

Answer: B

## Explanation

CBoW uses multiple context words to predict a target; skip-gram uses a target to predict multiple context words. Option A reverses them. Option C is wrong. Option D is wrong — both are unsupervised.

## Question

Cosine similarity between two word-embedding vectors measures which property?

- A. Euclidean distance.
- B. The cosine of the angle between the vectors, capturing semantic similarity independent of magnitude.
- C. Manhattan distance.
- D. Pearson correlation.

Answer: B

## Explanation

Cosine similarity = dot product divided by the product of magnitudes, i.e.  $\cos$  of the angle. It is robust to magnitude differences. Options A, C, and D are distinct metrics.

## Question

Which of the following best captures the architectural distinction between BERT and GPT?

- A. BERT is autoregressive; GPT is autoencoding.
- B. BERT uses an encoder-only architecture for bidirectional context (well-suited to understanding tasks); GPT uses a decoder-only autoregressive architecture for generation.
- C. Both are encoder-only.
- D. Both are decoder-only.

Answer: B

## Explanation

BERT is encoder-only, bidirectional (masked language modelling). GPT is decoder-only, autoregressive (next-token prediction). Option A reverses. Options C and D misclassify.

## Question

Although transformers have achieved state-of-the-art performance across NLP tasks, what is a key limitation?

- A. They cannot handle long sequences.
- B. They are computationally expensive and difficult to interpret due to their complex multi-layer architecture and self-attention patterns.
- C. They require labels for training.
- D. They cannot be fine-tuned.

Answer: B

## Explanation

Transformers are compute-intensive and opaque, complicating interpretability. Option A is wrong — they handle long sequences better than RNNs. Options C and D are incorrect (pretraining and fine-tuning are standard).